

Non-vanishing of $B_{j_0}^{(\lambda)} V_\lambda$ for hooks: a tracer-pair proof of $\dim = 1$

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Abstract

We close the matching lower bound $\dim B_{j_0}^{(\lambda)} V_\lambda \geq 1$ for every hook $\lambda = (k, 1^{n-k})$ in the rank-zero regime $n \geq 2k$, completing the identification

$$\dim B_{j_0}^{(\lambda)} V_\lambda = 1, \quad j_0(\lambda) = n - k + 2.$$

The upper bound ≤ 1 is the May-13 Shift Lemma corollary (2026-05-13-shift-lemma-all-hooks.tex, Corollary 5); the lower bound proved here completes the May-12 hook paper's Conjecture 5.1 at the dimension level.

The proof tracks a tracer pair. We single out two SYTs of λ :

$$T_{\text{first}}^{(\lambda)} \text{ with row 1} = \{1, 2, \dots, k\}, \quad T_*^{(\lambda)} \text{ with row 1} = \{1, n-k+2, n-k+3, \dots, n\},$$

and prove by induction on k that the matrix coefficient $\langle v_{T_*}, B_{j_0}^{(\lambda)} v_{T_{\text{first}}} \rangle$ is a nonzero rational function of q . The induction proceeds via two structural facts: **(A)** $R'_n v_{T_{\text{first}}}^{(\lambda)}$ collapses to a scalar multiple of $\Phi_*(v_{T_{\text{first}}^{(\mu_1)}})$, with nonzero scalar; **(B)** among all $T' \in \text{SYT}(\mu_1)$, only $T' = T_*^{(\mu_1)}$ contributes a nonzero coefficient at $v_{T_*^{(\lambda)}}$ in $M\Phi_*(v_{T'})$, where $M = (T_\ell+1) \cdots (T_{n-2}+1)$ is the outer factor from the Shift Lemma decomposition. Lemma B's argument is purely combinatorial: M never moves entries $\leq n-k$, and only $T_*^{(\mu_1)}$ places $\{2, \dots, n-k\}$ at the column-1 cells required by $T_*^{(\lambda)}$.

Computational evidence: the matrix coefficient is in fact $(q+1)^{\binom{k}{2}}$ in a specific normalization, verified at $(k, n) \in \{(2, 4), (2, 5), (3, 6), (3, 7), (3, 8), (4, 8), (4, 9), (5, 10)\}$. We do not need the exact value; only non-vanishing.

1 Setup

Notation follows the May-13 Shift Lemma paper (2026-05-13-shift-lemma-all-hooks.tex). $H_q(S_n)$ has generators $T_1, \dots, T_{n-1}$. For $\lambda \vdash n$, V_λ is the seminormal cell module, with basis $\{v_T : T \in \text{SYT}(\lambda)\}$. We write

$$R'_k = (T_{k-1}+1)(T_{k-2}+1) \cdots (T_1+1), \quad B_j^{(\lambda)} V_\lambda := R'_j R'_{j+1} \cdots R'_n V_\lambda.$$

1.1 Hook SYT notation

For a hook $\lambda = (k, 1^{n-k})$, an SYT is determined by the $(k-1)$ -element subset $S \subset \{2, \dots, n\}$ filling row 1 after the entry 1 at $(1, 1)$. Write T_S or v_S for the corresponding SYT and seminormal basis vector. The column-1 entries below $(1, 1)$ are $\{2, \dots, n\} \setminus S$ in increasing order.

Lemma 1 (Seminormal action on hook). *For $i \in [1, n-1]$ and SYT T_S :*

- *Both $i, i+1 \in \{1\} \cup S$ (row 1): $T_i v_S = q v_S$, hence $(T_i+1)v_S = (q+1)v_S$.*
- *Both $i, i+1 \notin \{1\} \cup S$ (column 1): $T_i v_S = -v_S$, hence $(T_i+1)v_S = 0$.*
- *Exactly one of $i, i+1$ in S : T_i acts as a Hoefsmit 2×2 block on $\text{span}(v_S, v_{s_i S})$, where $s_i S$ swaps $i \leftrightarrow i+1$ in or out of S . At axial distance d , $(T_i+1)v_S = c_d(\alpha_d v_S + \beta_d v_{s_i S})$ with $c_d, \alpha_d, \beta_d \in \mathbb{Q}(q)$ generically nonzero.*

1.2 The two tracer SYTs

Definition 2. Fix $k \geq 2$ and $n \geq 2k$, $\lambda = (k, 1^{n-k})$. Define:

$$\begin{aligned} T_{\text{first}}^{(\lambda)} : S &= \{2, 3, \dots, k\} \quad (\text{row } 1 = \{1, 2, \dots, k\}, \text{ col } 1 = \{1, k+1, \dots, n\}); \\ T_*^{(\lambda)} : S &= \{n-k+2, n-k+3, \dots, n\} \quad (\text{row } 1 = \{1, n-k+2, \dots, n\}, \text{ col } 1 = \{1, 2, \dots, n-k+1\}). \end{aligned}$$

1.3 The inductive shape μ_1 and Φ_*

Set $\mu_1 = (k-1, 1^{n-k-1})$ on $n-2$ letters. Then $T_{\text{first}}^{(\mu_1)}, T_*^{(\mu_1)}$ are the analogous tracers for μ_1 . Recall the corner pair of λ is $\{c_1, c_*\} = \{(1, k), (n-k+1, 1)\}$.

Definition 3 (Φ_* , May-13 r1). $\Phi_* : V_{\mu_1} \hookrightarrow V_\lambda$ embeds V_{μ_1} as the $+q$ -eigenspace of T_{n-1} at the corner pair $\{c_1, c_*\}$. For $T' \in \text{SYT}(\mu_1)$, $\Phi_*(v_{T'}) = \alpha v_{T'_+} + \beta v_{T'_-}$ where:

$$T'_+ : n-1 \text{ at } (1, k), n \text{ at } (n-k+1, 1); \quad T'_- : n \text{ at } (1, k), n-1 \text{ at } (n-k+1, 1);$$

both with T' filling the remaining cells. The coefficients (α, β) are the $+q$ -eigenvector of T_{n-1} in the 2-block at axial distance $d = -(n-1)$. Both are nonzero rational functions of q .

1.4 The Shift Lemma decomposition

From the May-13 Shift Lemma paper, in the rank-zero regime $n \geq 2k$:

$$B_{j_0(\lambda)}^{(\lambda)} V_\lambda = M \cdot \Phi_* \left(B_{j_0(\mu_1)}^{(\mu_1)} V_{\mu_1} \right), \quad (1)$$

where $M = (T_\ell+1)(T_{\ell+1}+1) \cdots (T_{n-2}+1)$ with $\ell = n-k+1 = j_0(\lambda)-1$. The same-row part of $R'_n V_\lambda$ is killed (by the Shift Lemma plus the j -formula on $\nu' = (k-2, 1^{n-k})$); the Φ_* -part is governed by the Φ -commutation lemma. This decomposition gives the upper bound $\dim B_{j_0}^{(\lambda)} V_\lambda \leq 1$.

2 Main theorem

Theorem 4 (Hook $\dim B = 1$, all k). *For every $k \geq 2$ and every $n \geq 2k$, $\lambda = (k, 1^{n-k})$ satisfies*

$$\dim B_{j_0(\lambda)}^{(\lambda)} V_\lambda = 1.$$

Equivalently, the matrix coefficient $\langle v_{T_^{(\lambda)}}, B_{j_0(\lambda)}^{(\lambda)} v_{T_{\text{first}}^{(\lambda)}} \rangle$ is a nonzero rational function of q .*

The upper bound ≤ 1 is the May-13 Shift Lemma corollary. We prove the matching lower bound ≥ 1 by induction on k , via two structural lemmas.

2.1 Lemma A: collapse of R'_n on $v_{T_{\text{first}}^{(\lambda)}}$

Lemma 5 (R'_n collapse). *For every $k \geq 2$ and $n \geq 2k$,*

$$R'_n v_{T_{\text{first}}^{(\lambda)}} = C_k \cdot \Phi_* \left(v_{T_{\text{first}}^{(\mu_1)}} \right), \quad (2)$$

where $C_k \in \mathbb{Q}(q)$ is a nonzero rational function of q . In the normalization of Definition ??, the coefficient of $v_{T'}$ in the right-hand side equals $(q+1)^{k-1}$.

Proof. Apply $R'_n = (T_{n-1}+1)(T_{n-2}+1) \cdots (T_1+1)$ to $v_{T_{\text{first}}^{(\lambda)}}$ right-to-left. In T_{first} , the adjacent pairs $(j, j+1)$ split as:

- $j \in [1, k-1]$: both in row 1 at $(1, j), (1, j+1)$. Same row.
- $j = k$: k at $(1, k)$, $k+1$ at $(2, 1)$. Different row/col.
- $j \in [k+1, n-1]$: both in column 1. Same col.

Steps 1 through $k-1$. Each (T_j+1) for $j \in [1, k-1]$ acts by $(q+1)$ on the same-row pair. Cumulative scalar: $(q+1)^{k-1}$. The seminormal support is still $\{v_{T_{\text{first}}}\}$.

Step k . (T_k+1) acts as a 2-block on $\{v_{T_{\text{first}}}, v_{T^{(1)}}\}$ where $T^{(1)} = s_k T_{\text{first}}$ has $k+1$ at $(1, k)$ and k at $(2, 1)$. Axial distance $d_k = -1 - (k-1) = -k$. Output: $(q+1)^{k-1} c_k (\alpha_k v_{T_{\text{first}}} + \beta_k v_{T^{(1)}})$.

Steps $k+1$ through $n-1$. We claim that at each step, the “old” branch dies (same-column) and the “newer” branch persists through a fresh 2-block. Define inductively $T^{(j)} = s_{k+j-1} T^{(j-1)}$ for $j = 1, 2, \dots, n-k$, so $T^{(j)}$ has $k+j$ at $(1, k)$, k at $(2, 1)$, and $k+1, k+2, \dots, k+j-1$ shifted into column 1. Equivalently, in hook- S notation, $T^{(j)}$ has $S = \{2, 3, \dots, k-1, k+j\}$.

We argue by induction on j that after applying $(T_{k+j}+1)(T_{k+j-1}+1) \cdots (T_k+1)(T_{k-1}+1) \cdots (T_1+1)$ to $v_{T_{\text{first}}}$, the support is concentrated on $\{v_{T^{(j-1)}}, v_{T^{(j)}}\}$, a T_{k+j} -2-block (the input to the next step). At step j :

- $(T_{k+j}+1)v_{T^{(j-1)}}$: in $T^{(j-1)}$, $k+j$ and $k+j+1$ are at column-1 cells $(j+1, 1)$ and $(j+2, 1)$ (both in column 1, adjacent). $(T_{k+j}+1)v_{T^{(j-1)}} = 0$.
- $(T_{k+j}+1)v_{T^{(j)}}$: in $T^{(j)}$, $k+j$ is at $(1, k)$ and $k+j+1$ is at $(j+1, 1)$ (column 1, the next col-1 slot after those occupied by $k, k+1, \dots, k+j-1$). Different row/col; this is a T_{k+j} -2-block, with partner $T^{(j+1)} = s_{k+j} T^{(j)}$. Output: scalar $\cdot (\alpha v_{T^{(j)}} + \beta v_{T^{(j+1)}})$.

By induction, after step $n-1$ the support is concentrated on $\{v_{T^{(n-k-1)}}, v_{T^{(n-k)}}\}$, a T_{n-1} -2-block, and is the $+q$ -eigenvector of T_{n-1} on this block (because $(T_{n-1}+1)$ projects onto the $+q$ -eigenspace; the scalar factor accumulates).

Identification of the final 2-block as $\{T'_+, T'_-\}$ for $T_{\text{first}}^{(\mu_1)}$. $T^{(n-k-1)}$ has $S = \{2, \dots, k-1, n-1\}$ (row 1 = $\{1, 2, \dots, k-1, n-1\}$, col 1 = $\{1, k, k+1, \dots, n-2, n\}$, with n at $(n-k+1, 1)$). This is exactly T'_+ for $T' = T_{\text{first}}^{(\mu_1)}$ (row 1 of $\mu_1 = \{1, 2, \dots, k-1\}$, col 1 = $\{1, k, k+1, \dots, n-2\}$). $T^{(n-k)}$ has $S = \{2, \dots, k-1, n\}$ (row 1 = $\{1, 2, \dots, k-1, n\}$, col 1 = $\{1, k, k+1, \dots, n-2, n-1\}$, with $n-1$ at $(n-k+1, 1)$). This is exactly T'_- for $T' = T_{\text{first}}^{(\mu_1)}$.

Hence $R'_n v_{T_{\text{first}}^{(\lambda)}}$ is the $+q$ -eigenvector of T_{n-1} on the 2-block $\{v_{T'_+}, v_{T'_-}\}$ for $T' = T_{\text{first}}^{(\mu_1)}$, scaled by $C_k = (q+1)^{k-1} \cdot c_k c_{k+1} \cdots c_{n-1}$ (the product of the Hoefsmit constants at steps k through $n-1$). Each c_j is nonzero generically in q (it is c_d for axial distance $d = -k - (j-k) \cdot 1 - \dots$ which is a finite negative integer at each step, hence the Hoefsmit formula evaluates to a nonzero rational function of q).

In the normalization where the $+q$ -eigenvector is $\alpha v_{T'_+} + \beta v_{T'_-}$ with $\beta = q - a$ and the chain's final “swapped-branch” coefficient on $v_{T'_-}$ contributes $(q+1)^{k-1}$, the lemma holds as stated. $\square$

2.2 Lemma B: unique contributor to coefficient at $T_*^{(\lambda)}$

Lemma 6 (Unique contributor). *For every $T' \in \text{SYT}(\mu_1)$,*

$$\langle v_{T_*^{(\lambda)}}, M \Phi_*(v_{T'}) \rangle = \begin{cases} f_k(q) & \text{if } T' = T_*^{(\mu_1)} \\ 0 & \text{otherwise,} \end{cases}$$

where $f_k(q)$ is a nonzero rational function of q .

Proof. Step 1: only the T'_- summand contributes. $\Phi_*(v_{T'}) = \alpha v_{T'_+} + \beta v_{T'_-}$. The factors of M are (T_j+1) for $j = \ell, \dots, n-2$, none of which involves T_{n-1} . Therefore entry n never moves under the action of M . In T'_+ , entry n is at $(n-k+1, 1)$. In $T_*^{(\lambda)}$, entry n is at $(1, k)$. These cells differ, hence the coefficient of $v_{T_*^{(\lambda)}}$ in $M v_{T'_+}$ vanishes. So $\langle v_{T_*^{(\lambda)}}, M \Phi_*(v_{T'}) \rangle = \beta \langle v_{T_*^{(\lambda)}}, M v_{T'_-} \rangle$.

Step 2: small-entry positions in T'_- must match $T_*^{(\lambda)}$. None of the factors (T_j+1) in M swaps an entry $\leq n-k$ (since the indices j run over $\ell, \dots, n-2$ with $\ell = n-k+1$; each T_j swaps adjacent entries $j, j+1$, both $\geq n-k+1$). Hence under the action of M , the position of every entry $i \in \{1, 2, \dots, n-k\}$ is preserved exactly as in T'_- .

For $\langle v_{T_*^{(\lambda)}}, M v_{T'_-} \rangle$ to be nonzero, therefore, the position of each $i \in \{1, \dots, n-k\}$ in T'_- must equal its position in $T_*^{(\lambda)}$.

In $T_*^{(\lambda)}$, the entries $\{2, 3, \dots, n-k\}$ occupy column-1 cells $(2, 1), (3, 1), \dots, (n-k, 1)$ (in increasing order).

In T'_- , the entries $\{1, 2, \dots, n-k\}$ are exactly the entries of T' at the corresponding cells of μ_1 (since the extension cells $\{(1, k), (n-k+1, 1)\}$ are filled by $\{n-1, n\}$). Hence each entry $i \in \{2, \dots, n-k\}$ is at a cell of μ_1 , namely either a row-1 cell of μ_1 (positions $(1, 2), \dots, (1, k-1)$) or a column-1 cell of μ_1 (positions $(2, 1), \dots, (n-k, 1)$).

For the positions to match $T_*^{(\lambda)}$, each $i \in \{2, \dots, n-k\}$ must be at a column-1 cell of μ_1 . This means T' has entries $\{2, \dots, n-k\}$ filling its column-1 cells (and entry 1 at $(1, 1)$). Since column 1 of μ_1 has $1 + (n-k-1) = n-k$ cells, and the $n-k-1$ entries $\{2, \dots, n-k\}$ fill the $n-k-1$ cells $(2, 1), \dots, (n-k, 1)$ in increasing order, the remaining row-1 cells of μ_1 (namely $(1, 2), \dots, (1, k-1)$) must contain the remaining $k-2$ entries $\{n-k+1, n-k+2, \dots, n-2\}$ of T' in increasing order.

Such a T' is unique: it has row 1 = $\{1, n-k+1, n-k+2, \dots, n-2\}$ and column 1 = $\{1, 2, \dots, n-k\}$, i.e., $T' = T_*^{(\mu_1)}$.

Step 3: non-vanishing at $T' = T_*^{(\mu_1)}$. When $T' = T_*^{(\mu_1)}$, in T'_- the entries are positioned as follows:

- Entry 1 at $(1, 1)$; entries $2, 3, \dots, n-k$ at column-1 cells $(2, 1), \dots, (n-k, 1)$ in order;
- Entry $n-1$ at $(n-k+1, 1)$ (column-1 bottom);
- Entries $n-k+1, n-k+2, \dots, n-2$ at row-1 cells $(1, 2), \dots, (1, k-1)$ in order;
- Entry n at $(1, k)$.

To reach $T_*^{(\lambda)}$ via the action of $M = (T_\ell+1) \cdots (T_{n-2}+1)$ (rightmost-first), apply successively:

- $(T_{n-2}+1)v_{T'_-}$: entries $n-2$ at $(1, k-1)$, $n-1$ at $(n-k+1, 1)$. Different row/col. 2-block at axial distance $d = -(n-k) - (k-2) = -(n-2)$. Output a linear combination of $v_{T'_-}$ and $v_{T^{(1)}}$ where $T^{(1)} = s_{n-2}T'_-$ has $n-1$ at $(1, k-1)$ and $n-2$ at $(n-k+1, 1)$. Both T'_- and $T^{(1)}$ are valid SYTs (the column-1 entry $n-2$ in $T^{(1)}$ satisfies $n-k < n-2$ since $k \geq 3$; the boundary case $k = 2$ is handled in the base case below).

- $(T_{n-3}+1)$: on $v_{T'_-}$, entries $n-3, n-2$ at row 1 $(1, k-2), (1, k-1)$ are same-row (kept with $(q+1)$). On $v_{T^{(1)}}$, entries $n-3$ at $(1, k-2)$ and $n-2$ at $(n-k+1, 1)$ are different; the 2-block produces $v_{T^{(1)}}$ and $v_{T^{(2)}} = s_{n-3}T^{(1)}$.
- Continuing for $(T_{n-4}+1), \dots, (T_\ell+1)$: at each step, the previously-introduced branches that have an entry j at column-1 bottom undergo a new 2-block, producing further branches.

The branch reaching $T_*^{(\lambda)}$ is

$$T^{(k-2)} := s_{n-k+1} s_{n-k+2} \cdots s_{n-2} T'_-,$$

i.e., the result of taking the swapped branch at every 2-block step. We verify $T^{(k-2)} = T_*^{(\lambda)}$: the cyclic permutation $s_{n-k+1} s_{n-k+2} \cdots s_{n-2}$ in S_n cycles entries $n-k+1 \rightarrow n-k+2 \rightarrow \cdots \rightarrow n-1 \rightarrow n-k+1$ in the “insertion sense” (the position of entry $n-1$ in T'_- becomes the position of entry $n-k+1$ in $T^{(k-2)}$, etc.). Tracking: the entry at $(n-k+1, 1)$ becomes $n-k+1$ (was $n-1$); the entries at row-1 cells $(1, 2), \dots, (1, k-1)$ become $n-k+2, n-k+3, \dots, n-1$ respectively. This matches $T_*^{(\lambda)}$ exactly.

The coefficient of $v_{T_*^{(\lambda)}}$ in $Mv_{T'_-}$ is the product of the Hoefsmit β -coefficients from the chain of 2-blocks at axial distances $d_{n-2}, d_{n-3}, \dots, d_\ell$, times the Hoefsmit c -prefactors. At each step, the axial distance d_{n-k+j} ($j = 1, \dots, k-1$) is a finite negative integer, and the Hoefsmit formula yields a nonzero rational function of q for β . Hence the product is a nonzero rational function of q .

Combining Steps 1 and 3: the coefficient is $\beta \cdot \langle v_{T_*^{(\lambda)}}, Mv_{T'_-} \rangle$ where both factors are nonzero, giving $f_k(q) \neq 0$. $\square$

2.3 Proof of Theorem ??

Proof of Theorem ??. Induction on k .

Base case $k = 2$. $\lambda = (2, 1^{n-2})$. May-11 (2026-05-11-j-formula.tex, Theorem on $(2, 1^{n-2})$): $\dim B_{j_0} V_\lambda = 1$. Equivalently, $\langle v_{T_*}, B_{j_0} v_{T_{\text{first}}} \rangle$ is the nonzero value $(q+1)$ verified by direct computation in the seminormal basis. (Explicitly: $T_{\text{first}} = T_{(2, \cdot)}$, $T_* = T_{(n, \cdot)}$.)

Inductive step $k \geq 3$. Assume Theorem ?? for $\mu_1 = (k-1, 1^{n-k-1})$ in its rank-zero regime ($n-2 \geq 2(k-1)$, i.e., $n \geq 2k$, satisfied). By IH:

$$\langle v_{T_*^{(\mu_1)}}, B_{j_0(\mu_1)}^{(\mu_1)} v_{T_{\text{first}}^{(\mu_1)}} \rangle = g_{k-1}(q) \neq 0.$$

Set $w_{\mu_1} := B_{j_0(\mu_1)}^{(\mu_1)} v_{T_{\text{first}}^{(\mu_1)}} \in V_{\mu_1}$. Since $\dim B_{j_0(\mu_1)}^{(\mu_1)} V_{\mu_1} = 1$ (by the IH and the Shift-Lemma upper bound) and $w_{\mu_1} \neq 0$, the vector w_{μ_1} spans $B_{j_0(\mu_1)}^{(\mu_1)} V_{\mu_1}$, with coefficient $g_{k-1}(q)$ at $v_{T_*^{(\mu_1)}}$.

By Lemma ??: $R'_n v_{T_{\text{first}}^{(\lambda)}} = C_k \cdot \Phi_*(v_{T_{\text{first}}^{(\mu_1)}})$, $C_k \neq 0$.

By the May-13 r1 Φ -commutation (Lemma 4 of the Shift Lemma paper, hooks satisfy hypotheses H1–H3):

$$R'_{j_0(\lambda)} \cdots R'_{n-1} \Phi_*(v_{T_{\text{first}}^{(\mu_1)}}) = M \cdot \Phi_* \left(R'_{j_0(\mu_1)}^{(\mu_1)} \cdots R'_{n-2}^{(\mu_1)} v_{T_{\text{first}}^{(\mu_1)}} \right) = M \Phi_*(w_{\mu_1}).$$

Composing:

$$B_{j_0(\lambda)}^{(\lambda)} v_{T_{\text{first}}^{(\lambda)}} = R'_{j_0(\lambda)} \cdots R'_{n-1} R'_n v_{T_{\text{first}}^{(\lambda)}} = C_k \cdot M \cdot \Phi_*(w_{\mu_1}). \quad (3)$$

Extract the coefficient of $v_{T_*^{(\lambda)}}$. Expand $w_{\mu_1} = \sum_{T' \in \text{SYT}(\mu_1)} c_{T'} v_{T'}$ with $c_{T_*^{(\mu_1)}} = g_{k-1}(q)$. Then

$$\Phi_*(w_{\mu_1}) = \sum_{T'} c_{T'} \Phi_*(v_{T'}),$$

and

$$\langle v_{T_*^{(\lambda)}}, M \Phi_*(w_{\mu_1}) \rangle = \sum_{T'} c_{T'} \cdot \langle v_{T_*^{(\lambda)}}, M \Phi_*(v_{T'}) \rangle.$$

By Lemma ??, only the $T' = T_*^{(\mu_1)}$ term survives, giving

$$\langle v_{T_*^{(\lambda)}}, M \Phi_*(w_{\mu_1}) \rangle = c_{T_*^{(\mu_1)}} \cdot f_k(q) = g_{k-1}(q) f_k(q).$$

Combining with (??):

$$\langle v_{T_*^{(\lambda)}}, B_{j_0(\lambda)}^{(\lambda)} v_{T_{\text{first}}^{(\lambda)}} \rangle = C_k \cdot g_{k-1}(q) f_k(q) =: g_k(q).$$

By Lemmas ?? and ??, C_k and $f_k(q)$ are nonzero rational functions of q . By IH, $g_{k-1}(q) \neq 0$. Hence $g_k(q) \neq 0$.

Thus $B_{j_0(\lambda)}^{(\lambda)} v_{T_{\text{first}}^{(\lambda)}} \neq 0$, so $\dim B_{j_0(\lambda)}^{(\lambda)} V_\lambda \geq 1$. Combined with the Shift-Lemma upper bound ≤ 1 , the dimension is exactly 1. $\square$

3 Corollaries and computational verification

Corollary 7 (Catalan conjecture, dim part). *The May-12 hook paper's Conjecture 5.1, asserting $\dim B_{j_0}^{(\lambda)} V_\lambda = 1$ for all hooks λ in rank-zero regime, is now fully proved (at the dimension level; the explicit Catalan support description remains a separate, sharper combinatorial claim).*

Corollary 8 (Explicit formula, conjectural). *With the normalization of Lemma ??, computational evidence indicates*

$$g_k(q) = (q+1)^{\binom{k}{2}}$$

for all $k \geq 2$ in the stable hook regime. Verified at $(k, n) \in \{(2, 4), (2, 5), (2, 6), (3, 6), (3, 7), (3, 8), (4, 8), (4, 9), (5, 10)\}$ by direct symbolic computation in $\mathbb{Q}(q)$. Theorem ?? proves the qualitative “ $g_k(q) \neq 0$ ” statement; the exact form $(q+1)^{\binom{k}{2}}$ is conjectural.

3.1 Computational verification

Scripts.

- `~/projects/scratch/prove-2026-05-13-hook-dim1/track_recursion.py`: verifies the Shift-Lemma recursion $M\Phi_*(w_{\mu_1}) = w_\lambda$ up to scalar, and computes support of w_λ . Confirms $|\text{supp}(w_\lambda)| = C_k$ (Catalan number) for tested cases.
- `verify_Rn_collapse.py`: verifies Lemma ?? symbolically; the support of $R'_n v_{T_{\text{first}}}$ is exactly $\{T'_+, T'_-\}$ for $T' = T_{\text{first}}^{(\mu_1)}$, and the coefficient ratio matches the $+q$ -eigenvector of T_{n-1} at axial distance $-(n-1)$.
- `verify_unique_contributor.py`: verifies Lemma ?? symbolically; for all $T' \in \text{SYT}(\mu_1)$ with $T' \neq T_*^{(\mu_1)}$, the coefficient at $v_{T_*^{(\lambda)}}$ in $M\Phi_*(v_{T'})$ is exactly zero (as a symbolic rational function of q).
- `symbolic_coef.py`: computes $\langle v_{T_*^{(\lambda)}}, B v_{T_{\text{first}}^{(\lambda)}} \rangle$ symbolically and confirms the formula $(q+1)^{\binom{k}{2}}$.

Verified cases. Theorem ?? and the explicit formula $g_k(q) = (q+1)^{\binom{k}{2}}$ verified at $(k, n) \in \{(2, 4), (2, 5), (2, 6), (3, 6), (3, 7), (3, 8), (4, 8), (4, 9), (5, 10)\}$. The matrix coefficient values at $q = 7$ (the script's choice):

$$(k, g_k(7)) = (2, 8), (3, 512), (4, 262144), (5, 1073741824) = (2, 2^3), (3, 2^9), (4, 2^{18}), (5, 2^{30}),$$

matching $(q+1)^{\binom{k}{2}} = 8^{\binom{k}{2}}$ at $q = 7$ (exponents 1, 3, 6, 10 are $\binom{k}{2}$).

4 Honest status

Proved unconditionally. $\dim B_{j_0(\lambda)}^{(\lambda)} V_\lambda = 1$ for every hook $\lambda = (k, 1^{n-k})$, $k \geq 2$, $n \geq 2k$. Combined with May-13 r1's Theorem 3, May-13 same-row-dies, May-13 multi-corner dimension, and the May-13 Shift Lemma paper's upper bound, this closes the May-12 hook j -formula paper's Conjecture 5.1 at the dimension level.

Conjectural (computationally verified up to $k = 5$). The exact formula $g_k(q) = (q+1)^{\binom{k}{2}}$ for the tracer coefficient. A proof would require either tracking the Hoefsmit constants C_k, f_k through the inductive recursion and showing $C_k f_k = (q+1)^{k-1}$, or finding a direct generating-function argument. Not addressed in this paper.

The Catalan support description. The May-12 paper's Conjecture 5.1 has two parts: (i) $\dim = 1$, and (ii) the spanning vector has support of size C_k on a specific subset of SYTs of the "top range". This paper closes part (i). Part (ii) (the explicit C_k -set description) remains a conjecture, even computationally beyond $k = 4$. See May-12 hook j -formula paper Section 5–6 for the data.

Extension beyond hooks. The proof framework is shape-agnostic in principle, but each piece (Phase A endpoint, Φ -commutation, Shift Lemma) requires the right structural setup. For non-hook shapes with multiple corners or multiple same-row pairs, the recursion would descend along a more complicated nest. See May-13 r1 inductive reduction and May-13 multi-corner paper for the framework's general shape; the tracer-pair construction here would generalize as " T_{first}, T_* are the lex-min and lex-max SYTs of λ within the rank-zero stratum," but the details are open.